

Case Study Rubric

Question 1 (1 point)

- Correctly identifies Prim's and Kruskal's algorithms as appropriate algorithms for this optimization problem. 0.4
- Avoids mentioning clearly inappropriate algorithms for this task, such as DFS or BFS as direct solutions to the optimization problem. 0.2
- Correctly explains Prim's algorithm as starting from a vertex and growing the minimum spanning tree by repeatedly selecting the minimum-weight adjacent edge. 0.15
- Correctly explains Kruskal's algorithm as constructing the minimum spanning tree by adding edges in increasing order of weight. 0.15
- Recognizes that cycles must not be formed in either approach. 0.05
- Correctly addresses the time complexity, avoiding incorrect claims such as stating that both algorithms are $O(n)$. 0.05

Question 2 (3 points)

- Correctly understands the objective of the problem: find the pair of pins whose shortest-path distance is maximum. 0.8
- Uses an appropriate shortest-path strategy to solve the problem. 0.7
- Correctly reconstructs and returns the intermediate pins forming the corresponding shortest path. 0.6
- Provides a coherent and functional implementation. 0.4
- Includes explanatory comments in the code. 0.2
- Includes an example of how to call the algorithm from a main program, specifying the initial parameter values. 0.3